Lesson 13: CPU to Apple Metal (MPS)

Companion diagram

Chip photo for device-compute context

Image source: https://commons.wikimedia.org/wiki/File:Mac_Mini_M1_chip.jpg

```
Goal
----
Move the exact same PyTorch workflow from CPU to Apple Metal when you are ready.

Device basics
-------------
A tensor and the model parameters must live on the same device.

Common devices:
- cpu
- cuda
- mps

Selection logic
---------------
The script checks devices in this order:

    if CUDA exists: use CUDA
    else if MPS exists: use MPS
    else: use CPU

Why start on CPU?
-----------------
CPU is simpler for learning:
- easier debugging
- fewer device-specific surprises
- easier printing and inspecting

When to switch to MPS
---------------------
```

```
Switch once you understand:
- tensors and shapes
- model forward pass
- training loop
- loss and backward pass

Diagram:

    CPU learning phase
           |
           v
    same code + different device
           |
           v
        MPS acceleration

Core rule
---------
Move both model and tensors:

    model.to(device)
    x = x.to(device)
    y = y.to(device)

If one stays on CPU and the other moves to MPS, PyTorch raises a device mismatch error.

What the script demonstrates
----------------------------
- selecting a device
- moving a model
- creating tensors on that device
- running forward pass
- computing loss
- running backward pass

Mental model
------------
This should feel like:

    same math
    same training loop
    different hardware target

What to try next
----------------
- Force CPU by setting device = torch.device("cpu")
- If MPS is available later, set device = torch.device("mps")
- Re-run earlier tutorials with the same device pattern

Self-check questions
--------------------
1. Why must model and tensors be on the same device?
2. Why is CPU a better learning environment at first?
3. What code usually changes when moving from CPU to MPS?
```

Read the matching Python file and run it while comparing the printed shapes, losses, and outputs against this handout.